Problem Set 33

Integration by Parts

1 Calculate the following integrals using integration by parts.

(a) $\int \arctan(x)\,dx$

Solution

Using integration by parts, let $u = \arctan(x)$ and $\frac{dv}{dx} = 1$. Then, $\frac{du}{dx} = \frac{1}{1+x^2}$ and $v = x$. So,

$$\int \arctan(x)\,dx = x\arctan(x) - \int \frac{x}{1+x^2}\,dx.$$

Now, to solve the integral $\int \frac{x}{1+x^2}\,dx$, we can use a simple substitution. Let $u = 1 + x^2$, then $du = 2x\,dx$. The integral then simplifies to

$$\frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln|u| + C = \frac{1}{2}\ln(1+x^2) + C.$$

Therefore, the solution to the original integral is

$$x\arctan(x) - \frac{1}{2}\ln(1+x^2) + C.$$

(b) $\int x\cos(x)\,dx$

Solution

Set $u = x$ and $\frac{dv}{dx} = \cos(x)$. Then, $\frac{du}{dx} = 1$ and $v = \sin(x)$. Therefore,

$$\int x\cos(x)\,dx = x\sin(x) - \int \sin(x)\,dx = x\sin(x) + \cos(x) + C.$$

(c) $\int (\ln x)^2\,dx$

Solution

Let $u = (\ln x)^2$ and $\frac{dv}{dx} = 1$. Thus, $\frac{du}{dx} = 2\ln x\frac{1}{x}$ and $v = x$. The integral becomes

$$x(\ln x)^2 - 2\int x\ln x\frac{1}{x}dx = x(\ln x)^2 - 2\int \ln x\,dx.$$

Now, we solve the integral $\int \ln x\,dx$ again by integration by parts. Let $u = \ln x$ and $\frac{dv}{dx} = 1$, then $\frac{du}{dx} = \frac{1}{x}$ and $v = x$. Thus,

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - \int 1\,dx = x\ln x - x.$$

Substituting this back into our original equation, we get

$$x(\ln x)^2 - 2(x\ln x - x) = x(\ln x)^2 - 2x\ln x + 2x.$$

Thus, the solution to the integral $\int (\ln x)^2\,dx$ is

$$x(\ln x)^2 - 2x\ln x + 2x + C,$$

(d) $\displaystyle\int e^x \sin(x)\,dx$

Solution

First let $u = e^x$ and $dv = \sin(x)dx$. Then, $du = e^x dx$ and $v = -\cos(x)$. Thus,

$$\int e^x \sin(x)\,dx = -e^x\cos(x) + \int e^x\cos(x)\,dx.$$

The remaining integral $\int e^x\cos(x)\,dx$ we also need to solve by parts! Once again, let's take $u = e^x$ and $dv = \cos x\ dx$ so that $du = e^x dx$ and $v = \sin x$. Then

$$\int e^x\cos(x)\,dx = e^x\sin(x) - \int e^x\sin(x)\,dx.$$

Putting this together, we have

$$\int e^x\sin(x)\,dx = -e^x\cos(x) + e^x\sin(x) - \int e^x\sin(x)\,dx.$$

Now, since the same indefinite integral appears on both sides, we can move it all to the left and solve for it. Notice that because all indefinite integrals have an implicit " $+C$", once we move all indefinite integrals from the right side, we must include this term on the right:

$$2\int e^x\sin(x)\,dx = -e^x\cos(x) + e^x\sin(x) + C$$

or finally,

$$\int e^x\sin(x)\,dx = \frac{e^x(\sin x - \cos x)}{2} + C$$

(e) $\displaystyle\int x^2\sin(x)\,dx$

Solution

Using integration by parts twice, first let $u = x^2$ and $dv = \sin(x)dx$. Then,

$$\int x^2\sin(x)\,dx = -x^2\cos(x) + 2\int x\cos(x)\,dx.$$

The remaining integral $\int x\cos(x)\,dx$ is solved as in part (b). So,

$$\int x^2\sin(x)\,dx = -x^2\cos(x) + 2(x\sin(x) + \cos(x)) + C$$

2 Calculate the following integrals by using an appropriate strategy.

(a) $\displaystyle\int x\cos(x^2)\,dx$.

Solution

Use substitution, setting $u = x^2$. Then, $du = 2xdx$. The integral becomes

$$\frac{1}{2}\int \cos(u)\,du = \frac{1}{2}\sin(u) + C = \frac{1}{2}\sin(x^2) + C.$$

(b) $\displaystyle\int_1^2 \frac{2xe^{x^2}}{e^{x^2+5}}\,dx$

Solution

Simplify the integral:

$$\int_1^2 \frac{2xe^{x^2}}{e^{x^2+5}}\,dx = \int_1^2 2xe^{-5}\,dx.$$

Now, we can integrate $2x$ with respect to x:

$$\int 2x\,dx = x^2.$$

And we can calculate:

$$e^{-5}\left[x^2\right]_1^2 = e^{-5}\left(2^2 - 1^2\right) = e^{-5}(4-1) = 3e^{-5}.$$

Therefore, the solution to the integral is

$$\int_1^2 \frac{2xe^{x^2}}{e^{x^2+5}}\,dx = 3e^{-5}.$$

(c) $\displaystyle\int x\ln\left(x^2+1\right)\,dx$.

Solution

Use substitution, setting $u = x^2 + 1$. Then, $du = 2x\,dx$. The integral becomes

$$\frac{1}{2}\int \ln(u)\,du.$$

To solve this integral using integration by parts, let $v = \ln(u)$ and $dw = du$. Then, $dv = \frac{1}{u}\,du$ and $w = u$. Integration by parts gives us

$$\int \ln(u)\,du = u\ln(u) - \int u\cdot\frac{1}{u}\,du = u\ln(u) - \int 1\,du = u\ln(u) - u.$$

Substituting back $u = x^2 + 1$, the integral becomes

$$\frac{1}{2}\left[(x^2+1)\ln(x^2+1) - (x^2+1)\right] + C.$$

Therefore, the solution to the original integral $\int x\ln(x^2+1)\,dx$ is

$$\frac{1}{2}\left[(x^2+1)\ln(x^2+1) - (x^2+1)\right] + C$$

(d) $\displaystyle\int \frac{3x^2}{x^2+2x+1}\,dx.$

Solution

First, factor the denominator:

$$x^2 + 2x + 1 = (x+1)^2.$$

So the integral becomes

$$\int \frac{3x^2}{(x+1)^2}\,dx.$$

Next, use the substitution $u = x + 1$, which implies $du = dx$ and $x = u - 1$. Substituting these into the integral gives:

$$\int \frac{3(u-1)^2}{u^2}\,du.$$

Expanding the numerator:

$$\int \frac{3(u^2-2u+1)}{u^2}\,du = \int 3\left(1 - \frac{2}{u} + \frac{1}{u^2}\right)du.$$

Now, integrate term by term:

$$3\int 1\,du - 6\int \frac{1}{u}\,du + 3\int \frac{1}{u^2}\,du = 3u - 6\ln|u| - \frac{3}{u} + C.$$

Finally, substitute back $u = x + 1$:

$$3(x+1) - 6\ln|x+1| - \frac{3}{x+1} + C.$$

Thus, the solution to the integral $\int \frac{3x^2}{x^2+2x+1}\,dx$ is

$$3(x+1) - 6\ln|x+1| - \frac{3}{x+1} + C.$$

(e) $\displaystyle\int x^3 e^{x^2}\,dx.$

Solution

Use the substitution method, setting $u = x^2$. Then, $du = 2x\,dx$, and thus $x\,dx = \frac{1}{2}\,du$. The integral becomes

$$\int x^3 e^{x^2}\,dx = \int x^2 \cdot x e^{x^2}\,dx.$$

Substituting $u = x^2$ and $x\,dx = \frac{1}{2}\,du$, we get

$$\int x^2 e^u \cdot \frac{1}{2}\,du.$$

Now, replace x^2 with u:

$$\frac{1}{2}\int u e^u\,du.$$

This integral can be solved using integration by parts. Let $v = u$ and $\frac{dw}{du} = e^u$. Then $\frac{dv}{du} = 1$ and $w = e^u$. Applying integration by parts gives us

$$\frac{1}{2}\left(ue^u - \int e^u\,du\right) = \frac{1}{2}\left(ue^u - e^u\right) + C.$$

Substituting back $u = x^2$, we get

$$\frac{1}{2}\left(x^2 e^{x^2} - e^{x^2}\right) + C.$$

Therefore, the solution to the integral $\int x^3 e^{x^2}\,dx$ is

$$\frac{1}{2}\left(x^2 e^{x^2} - e^{x^2}\right) + C.$$

(f) $\displaystyle\int -\frac{\ln(1+x)}{(1+x)^2}\,dx.$

Solution

First, let's employ the substitution $u = 1 + x$, which implies $du = dx$. The integral becomes

$$\int -\frac{\ln(u)}{u^2}\,du.$$

Now, we apply integration by parts. Let's choose $v = \ln(u)$ and $\frac{dw}{du} = -\frac{1}{u^2}$. Therefore, $\frac{dv}{du} = \frac{1}{u}$ and $w = \frac{1}{u}$. The integration by parts formula gives us

$$\int -\frac{\ln(u)}{u^2}\,du = \frac{\ln(u)}{u} - \int \frac{1}{u}\cdot\frac{1}{u}\,du = \frac{\ln(u)}{u} - \int \frac{1}{u^2}\,du.$$

The integral $\int \frac{1}{u^2}\,du$ is straightforward:

$$\int \frac{1}{u^2}\,du = -\frac{1}{u}.$$

Substituting this back, we get

$$\frac{\ln(u)}{u} + \frac{1}{u} + C = \frac{\ln(u) + 1}{u} + C.$$

Finally, substituting back $u = 1 + x$, the integral becomes

$$\frac{\ln(1+x) + 1}{1+x} + C.$$

Therefore, the solution to the integral $\int -\frac{\ln(1+x)}{(1+x)^2}\,dx$ is

$$\frac{\ln(1+x) + 1}{1+x} + C.$$